Retrieval Quality and RAG Evaluation

Getting the Right Evidence Before Generating the Answer

Module 2 — Lecture 3

MODULE 2 · LECTURE 3 | DURATION · 60 MIN | SYLLABUS · 2.1 INTRODUCTION TO RAG

TOPICS COVERED

- Similarity Search
- Cosine Similarity
- Chunking Strategies
- Chunk Size & Overlap
- Top-k
- Metadata Filtering
- Reranking
- Hybrid Retrieval
- Hallucination Reduction
- Precision@k · Recall@k · MRR



COURSE ROADMAP MODULE 2 · LECTURE 3 OF 4

Where Are We? From Engine to Reliability

LECTURE 1

Why RAG?

External knowledge + LLM

The motivation

→

LECTURE 2

How RAG Works

Load → Chunk → Embed → Store
→ Retrieve → Augment → Generate

The engine

→

LECTURE 3 · TODAY

How Do We Make RAG Reliable?

Similarity | Chunking | Top-k | Metadata
Reranking | Hallucination Control | Evaluation

The quality layer

→

LECTURE 4

Real-World Applications

Where reliable RAG is deployed

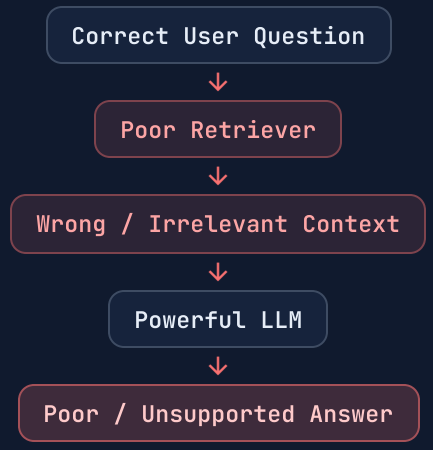
Next stop

Key message: Lecture 2 built the RAG engine. **Lecture 3 improves the quality of what that engine retrieves.**

THE CORE PROBLEM · SAME QUESTION · SAME LLM · DIFFERENT RETRIEVAL

The Fundamental RAG Quality Problem

✕ POOR RETRIEVER



✓ GOOD RETRIEVER



FAILURE TYPE 1 — RETRIEVAL FAILURE

The correct evidence **was not retrieved**.

`Question → Wrong chunks`

FAILURE TYPE 2 — GENERATION FAILURE

Correct evidence was retrieved, but the LLM **misused it**.

`Correct chunks → Incorrect answer`

Diagnostic question: "Was the evidence wrong, or did the model use correct evidence incorrectly?"

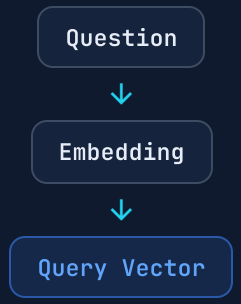
📌 **Memory anchor:** `Garbage retrieved → garbage augmented → garbage generated.` Learn to separate the two failure types.

PART A — SIMILARITY SEARCH

VECTORS ON BOTH SIDES OF THE SEARCH

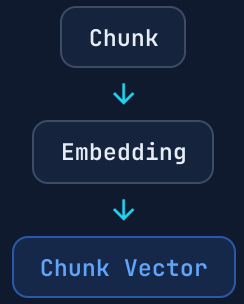
What Is Similarity Search?

RECALL LECTURE 2 — THE QUERY SIDE



The user's question becomes a **point in vector space**.

STORED CHUNKS ALSO HAVE VECTORS



Every indexed chunk is **also a point** in the same space.

NOW WE ASK

Which stored vectors are **most similar** to the query vector?

That is similarity search. Retrieval ranks stored chunks by vector similarity to the query — not by exact wording.

PART A — SIMILARITY SEARCH SAME MEANING · DIFFERENT WORDS

Semantic Similarity in Action

USER ASKS *"My semester fee was deducted but the portal still shows unpaid."*

CHUNK A ✓ "Debited fee transactions not reflected on the portal require **payment reconciliation**."

CHUNK B "The central library is open until 8 PM."

CHUNK C "Students may apply for transcripts online."

Which is most semantically related? Chunk A — even though the wording is different.

Takeaway: semantic retrieval matches **meaning**, not merely identical words.

PART A — SIMILARITY SEARCH TWO WAYS TO MATCH · NEITHER WINS EVERYWHERE

Keyword Search vs Semantic Search

KEYWORD / LEXICAL MATCHING	SEMANTIC RETRIEVAL
Looks for **term overlap**	Looks for **embedding similarity**
Strong when **exact terminology** matters	Strong when **wording differs**
`"payment failed"`	`"transaction unsuccessful"`
Can **miss synonyms**	Can capture **semantic similarity**
Easy to explain	Depends on **embedding quality**

⚠ **Important:** semantic search does **not** completely replace keyword search — **later we will combine them** (hybrid retrieval, Part F).

PART A — SIMILARITY SEARCH BLACKBOARD ITEM — ONE FORMULA, ONE PICTURE

Cosine Similarity — Formula and Intuition

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \, \|B\|}$$

WHERE

- A = query vector
- B = document / chunk vector
- $A \cdot B$ = dot product
- $\|A\|, \|B\|$ = vector magnitudes

Basic intuition: cosine similarity measures **how similarly two vectors point**. More similar direction → higher score; less similar direction → lower score.

$A = [1, 0]$ $B = [1, 0]$

Same direction:

$\cos(\theta) = 1$

Very similar.

$C = [0, 1]$

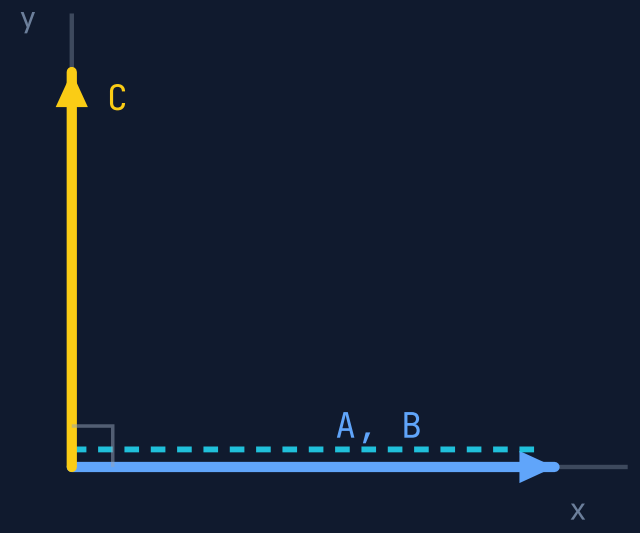
Perpendicular to A:

$\cos(\theta) = 0$

Less similar.

Range: for general vectors cosine similarity runs from -1 to 1, although practical embedding distributions vary.

THE PICTURE



A and B point the same way. C is perpendicular — **direction, not length, decides the score**.

PART A — SIMILARITY SEARCH THE MOST IMPORTANT CAVEAT OF THE HOUR

Why Cosine? And Similarity Is Not Correctness

WHY COSINE AND NOT RAW DISTANCE?

```
Vector A = [10, 0]
```

```
Vector B = [1, 0]
```

Magnitudes differ — 10× apart — but **directions are identical**.

Cosine similarity focuses mainly on **direction** rather than absolute vector magnitude.

Memory sentence: cosine similarity asks — *"Are these vectors pointing in a similar semantic direction?"*

VERY IMPORTANT — SIMILARITY ≠ CORRECTNESS

Query:

*"What is the **2026** examination deadline?"*

A chunk about:

*"**2025** examination deadline"*

may be **highly semantically similar** — but it is still the **wrong answer source**.

Therefore: high embedding similarity does not automatically mean factual relevance.

This motivates:

Metadata | Recency | Reranking | Filtering

PART B — CHUNKING QUALITY FROM "WHAT" TO "HOW"

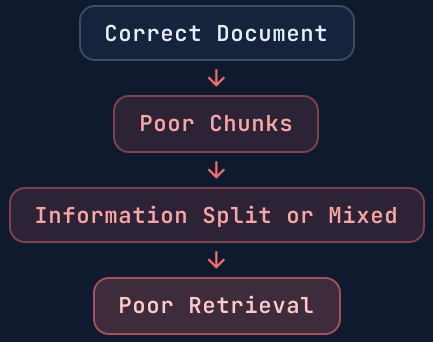
Why Revisit Chunking? Strategy 1 — Fixed-Size

THE QUESTION CHANGES

Lecture 2 asked: "What is chunking?"

Lecture 3 asks: "How *should* we chunk?"

Because poor chunking can create:



Chunking is therefore a **retrieval-quality decision**.

1 Fixed-Size Chunking

Split every fixed number of **characters, words, or tokens**.

Chunk 1 → tokens 1–300

Chunk 2 → tokens 251–550 (overlap 49)

Chunk 3 → tokens 501–800 (overlap 49)

ADVANTAGES

Simple · predictable · easy to implement

LIMITATION

May split text at unnatural semantic boundaries

PART B — CHUNKING QUALITY STRATEGIES 2 & 3

Sentence / Paragraph and Recursive Chunking

2 Sentence / Paragraph Chunking

Split using **natural text boundaries**:

```
Paragraph 1 → Chunk 1
Paragraph 2 → Chunk 2
Paragraph 3 → Chunk 3
```

ADVANTAGES

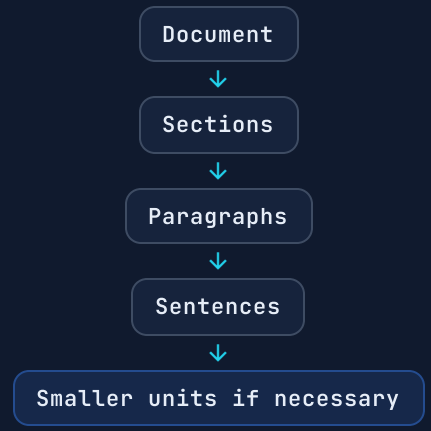
Preserves natural meaning · easier to interpret

LIMITATION

Paragraph sizes may vary significantly

3 Recursive Chunking

Try larger natural separators **first**:



Goal: preserve natural structure while remaining within a target chunk size. Common in practical RAG frameworks.

PART B — CHUNKING QUALITY STRATEGIES 4 & 5

Semantic and Structure-Aware Chunking

4 Semantic Chunking

Instead of splitting only by length: **break when the topic / meaning changes**.

Paragraphs 1–3	Admissions
Paragraphs 4–7	Fee Payment
Paragraphs 8–11	Examination
Paragraphs 12–15	Attendance

Each semantic topic can form a chunk.

ADVANTAGE

Potentially more coherent retrieval units

DISADVANTAGE

More computation / complexity

5 Structure-Aware Chunking

For **structured documents**, these can be natural retrieval units:

Chapter · Section · Subsection · Table · FAQ Question · Policy Clause

EXAMPLE — ONE LOGICAL CHUNK

Section 4.2 — Examination Registration

This can be **better than splitting every arbitrary 500 tokens** — the document's own organization is preserved.

PART B — CHUNKING QUALITY FIVE STRATEGIES · SIDE BY SIDE

Chunking Strategy Comparison

STRATEGY	STRENGTH	LIMITATION
Fixed-size	Simple	Can split meaning
Paragraph	Natural boundaries	Inconsistent sizes
Recursive	Practical balance	More logic
Semantic	Meaning-aware	More expensive
Structure-aware	Preserves document organization	Depends on document structure

Key point: there is **no universally best chunking strategy**. Ask the class: is one chunking strategy universally best? The honest answer decides how you engineer each system.

PART B — CHUNKING QUALITY · TWO DIALS · SIZE AND OVERLAP

Chunk Size and Overlap Trade-offs

DIAL 1 — CHUNK SIZE

TOO SMALL

`Chunk: "The deadline is..." · separate chunk: "18 October."`

Problem: meaning may be split.

TOO LARGE

One chunk contains: Admissions · Fees · Examination · Attendance · Library

Problem: retrieval becomes less focused.

Memory anchor: small chunks improve **precision** but may lose context. Large chunks preserve **context** but may reduce specificity.

DIAL 2 — CHUNK OVERLAP

CHUNK SIZE	OVERLAP
500 tokens	100 tokens

Overlap helps **preserve boundary information**. But:

More overlap → More duplicated text → More embeddings → More storage → Potential repeated retrieval

Therefore: overlap should preserve context **without creating unnecessary duplication**.

PART C — TOP-K RETRIEVAL BLACKBOARD ITEM — ONE NUMBER CONTROLS THE FUNNEL

What Is Top-k?

THE EXAMPLE

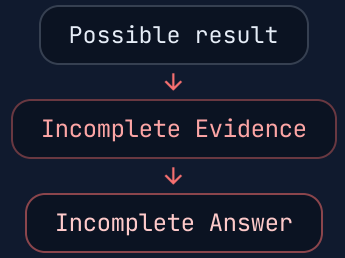
Suppose the retriever **scores 100 chunks**. If `k = 3`, we return the **three highest-ranked chunks**:

- RANK 1 — Chunk A
- RANK 2 — Chunk B
- RANK 3 — Chunk C

These are **passed to the next stage** — augmented into the prompt.

WHAT IF k IS TOO SMALL? — k = 1

The question needs information from **two different sections**. The retriever returns **only one**.



Risk: important evidence may be missed — the model never even sees it.

Blackboard: `Top-k = retrieve the k highest-ranked candidates.` Ask the class: what does Top-k mean? One number, a whole behavior.

PART C — TOP-K RETRIEVAL MORE IS NOT AUTOMATICALLY BETTER

The Top-k Trade-off

WHAT IF k IS TOO LARGE? — k = 20 Now the context contains:

relevant chunks | weakly relevant chunks | possibly contradictory chunks | repeated content

more tokens | more latency | more cost | more distraction | possibly poorer generation

Small k

→ focused context
→ but may miss evidence

Large k

→ better chance of including evidence
→ but more noise and cost

Goal: retrieve **enough evidence to answer the question** without **overwhelming the generator**. Key point: **more context is not automatically better context**.

PART D — METADATA FILTERING · THE SECOND SIGNAL BESIDE SIMILARITY

Metadata Filtering

Query: *"What is the 2026 Semester VIII deadline?"* — vector retrieval may find all of these, **all semantically very similar**:

- NEED — 2026 Semester VIII Circular
- WRONG — 2025 Semester VIII Circular
- WRONG — 2024 Semester VIII Circular

We need another signal: **Metadata**.

EACH CHUNK MIGHT HAVE

`year = 2026` `department = Examination` `document_type = Circular` `course = B.Tech` `semester = VIII`

Query `"2026 Semester VIII examination deadline"` can apply `year = 2026` `semester = VIII` `department = Examination` — before or along with semantic retrieval.

BENEFIT

Filtering **removes obviously irrelevant candidates before ranking**.

USEFUL METADATA CAN INCLUDE

date · department · document type · author · product · region · access level · version · course · academic year

PRODUCTION BENEFIT — ACCESS CONTROL

Metadata also enforces permissions: **HR documents should not be retrieved for unauthorized users.** An important enterprise point.

PART E — RERANKING RETRIEVE BROADLY · RERANK CAREFULLY

Why Rerank? What Is a Reranker?

Why rerank? Vector search is designed to retrieve candidates **efficiently** — but the top result is not always the most relevant.

Question → Vector Retrieval → Top 20 Candidates → RERANKER → Best 5 Candidates → LLM

WHAT A RERANKER DOES

Evaluates the relationship between **Question + Candidate Chunk** more carefully than the initial retrieval stage:

- Query + Chunk A → relevance score
- Query + Chunk B → relevance score

Then **reorder the candidates**.

POSSIBLE RERANKERS INCLUDE

- Cross-encoder models
- Specialized reranking models
- LLM-based reranking in some systems

Keep implementation details for later work.

Blackboard: `Retrieve → Rerank → Generate.` Memory anchor: **retrieve broadly; rerank carefully.**

PART E — RERANKING · TWO STAGES · TWO JOBS

Retriever vs Reranker

RETRIEVER	RERANKER
Searches **large corpus**	Works on **small candidate set**
Optimized for **speed**	Optimized for **relevance**
Returns candidates	**Reorders** candidates
Often embedding-based	May **jointly consider** query + chunk
First stage	**Second** stage

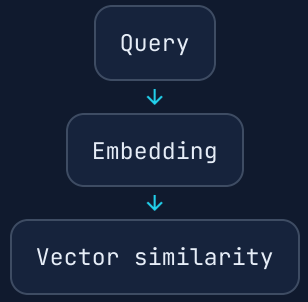
Key idea: `Fast retrieval first, expensive precision second.` Ask the class: what is the difference between a retriever and a reranker? Stage, scope, and objective.

PART F — DENSE · LEXICAL · HYBRID MEANING AND EXACT TERMS, COMBINED

Dense, Lexical and Hybrid Retrieval

Dense Retrieval

Uses **embeddings**.



Strength: semantic matching.

```
"payment failed"
≈ "transaction unsuccessful"
```

Lexical Retrieval

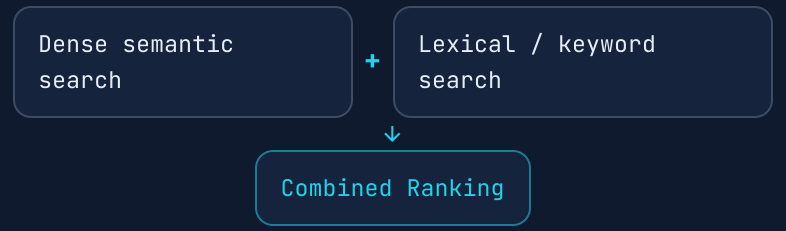
Uses **words / terms** — e.g. `inverted index`, `BM25`.

Strong when exact terms matter.

```
Circular CE/EXAM/2026/147
```

Exact identifier matching may be very important — semantic embeddings alone may not always be ideal.

Hybrid Retrieval = `combine`



Dense finds **similar meaning**. Lexical finds **exact terminology**. Together they complement each other.

```
Query: "Policy CE/EXAM/2026/147 late examination form" — exact policy number + semantic meaning both matter.
```

PART G — HALLUCINATION REDUCTION REDUCE, NOT ELIMINATE

Does RAG Eliminate Hallucination? No.

POSSIBLE FAILURE — PATH 1

Correct Retrieved Context
↓ LLM
Adds Unsupported Detail

POSSIBLE FAILURE — PATH 2

Wrong Retrieval
↓ Model confidently answers

CORRECT STATEMENT

RAG can **reduce** hallucination by grounding generation in evidence, but it **cannot guarantee hallucination-free answers**.

WAYS TO REDUCE UNSUPPORTED GENERATION

1. **Improve retrieval** — better evidence.
2. **Rerank** — use the most relevant context.
3. **Give explicit grounding instructions** — "Use only the supplied context."
4. **Allow abstention** — "If the context does not contain the answer, say that the information is unavailable."
5. **Return sources** — evidence becomes inspectable.
6. **Validate output where appropriate** — structured checks / business rules.

PART G — HALLUCINATION REDUCTION PROMPT DESIGN + SHOW YOUR SOURCES

Grounded Prompt and Source Attribution

SYSTEM:

Answer only from the supplied context. If the answer cannot be found, state:
"I do not have enough information in the provided documents."

CONTEXT:

[retrieved chunks]

QUESTION:

What penalty applies after the deadline?

This is better than: "Answer the question." Prompting **cannot repair completely wrong retrieval**, but it can encourage the generator to **respect retrieved evidence**.

A USEFUL RAG ANSWER MAY CONTAIN

Answer: The deadline is 18 October 2026.

Source: Examination Circular 2026, Page 4.

BENEFITS

- traceability
- verification
- trust
- debugging

But important: the system should ensure the cited source actually supports the claim. **Citation appearance alone is not proof.**

PART H — RAG EVALUATION SPLIT THE PIPELINE BEFORE YOU SCORE IT

How Do We Evaluate RAG?

DO NOT EVALUATE ONLY THE FINAL ANSWER



Why split? A wrong answer may be caused by **retrieval, generation, or both**.

RETRIEVAL EVALUATION QUESTION — FOR EVERY TEST QUERY

"Did the retriever return the evidence required to answer the question?"

Question: *"What is the examination deadline?"* — relevant chunk: `Exam_Circular_Page4`. Retrieved Top-3:

1. Fee Policy
2. Exam Circular Page 4 ✓ **relevant**
3. Hall Ticket Rules

The relevant chunk was retrieved. **That's good retrieval coverage.**

PART H — RAG EVALUATION BLACKBOARD ITEM — THREE BEGINNER METRICS

Retrieval Metrics: Precision@k · Recall@k · MRR

Precision@k

Top-5 results contain `3 relevant` + `2 irrelevant`:

$$\text{Precision@5} = 3/5 = 0.60$$

Intuition: of what we retrieved, how much was relevant?

High precision → little irrelevant context.

Recall@k

`4 total` relevant chunks exist; Top-5 retrieves `3 of them`:

$$\text{Recall@5} = 3/4 = 0.75$$

Intuition: of all relevant evidence, how much did we successfully retrieve?

High recall → few relevant chunks missed.

MRR — Mean Reciprocal Rank

First relevant result appears at:

Rank 1	RR = 1
Rank 2	RR = 1/2
Rank 5	RR = 1/5

Average these reciprocal ranks over many queries → MRR.

How early does the first relevant result appear? Higher is better.

⚖ **Trade-off:** increase k → recall may improve, but precision may fall. Not guaranteed in every case — useful intuition. Top-k often drives this balance.

PART H — RAG EVALUATION NOW ASSUME EVIDENCE WAS RETRIEVED

Generation Evaluation and Correctness vs Faithfulness

FIVE QUESTIONS FOR THE GENERATED ANSWER

- **Answer Correctness** — is the answer correct?
- **Faithfulness / Groundedness** — does it actually use / support itself from the context?
- **Answer Relevance** — does it answer the user's question?
- **Citation Accuracy** — are citations correct?
- **Format Compliance** — did it follow the required format?

Blackboard: Generation — Correctness · Faithfulness · Relevance.

THESE ARE DIFFERENT — WORKED EXAMPLE

Context says:

```
"Deadline: 18 October."
```

Model answers:

"Deadline is 18 October **and a ₹500 late fee applies**."

Maybe the deadline is correct — but `₹500` was **not in the context**. Correct component + **unsupported component** → faithfulness is weaker.

Key point: a response can contain correct information and still include unsupported claims. Ask: can an answer be correct but not faithful?

PART H — RAG EVALUATION

ACCURATE AND OPERATIONALLY RELIABLE

End-to-End Evaluation — Quality Is More Than Accuracy

A PRODUCTION EVALUATION SET MIGHT CONTAIN

QUESTION	EXPECTED ANSWER	RELEVANT SOURCE
Exam deadline?	18 Oct	Exam Circular
Late form?	Approval required	Exam Circular
Refund process?	As policy	Fee Policy

FOR EACH QUERY, EVALUATE

Retrieved correct evidence? → Answer correct? → Answer grounded?

Source correct? → Latency acceptable?

PRODUCTION RAG MAY NEED TO MEASURE

RETRIEVAL

Precision@k · Recall@k · MRR

GENERATION

correctness · relevance · faithfulness · citation quality

SYSTEM

latency · memory · throughput · cost

SAFETY / SECURITY

permissions · private-data leakage · inappropriate retrieval

Key point: a good RAG system is both **accurate** and **operationally reliable**.

DEBUGGING RAG AN EXCELLENT DEBUGGING MAP — KEEP IT

RAG Failure Taxonomy and Seven Debugging Questions

RAG FAILURE — SEVEN PLACES IT BREAKS

- **Data Failure** — bad / outdated documents
- **Chunking Failure** — evidence split badly
- **Embedding Failure** — meaning represented poorly
- **Retrieval Failure** — correct chunk not returned
- **Ranking Failure** — good chunk ranked too low
- **Context Failure** — too much / conflicting context
- **Generation Failure** — LLM misuses evidence

WHEN A RAG ANSWER IS WRONG, ASK:

1. Is the correct information in the knowledge base?
2. Was the document processed correctly?
3. Was the evidence chunked correctly?
4. Did retrieval return the correct chunk?
5. Was the correct chunk ranked high enough?
6. Was useful context passed to the LLM?
7. Did the LLM use the context faithfully?

📌 **Memory anchor:** `Debug RAG from left to right.` Do not immediately blame the LLM. Ask: when the final answer is wrong, what should you inspect first?

WORKED EXAMPLE → YOUR TURN DEMONSTRATE, THEN PRACTICE

Worked University Example and Classroom Activity

DEMO — COMPLETE QUALITY PIPELINE

"Can I submit my Semester VIII examination form after 18 October?"

- A Deadline is 18 October.
- B Late submissions require Examination Cell approval.
- C Library books must be returned before examination.

SIMILARITY RETURNS

1. A · 2. C · 3. B — evidence present, ranking imperfect

RERANKING PRODUCES

1. A · 2. B · 3. C — better context

LLM answers: *"Yes, but late submission requires approval from the Examination Cell."*

ACTIVITY — YOU GRADE THIS RETRIEVAL

"What should I do if my fee was deducted but still shows unpaid?"

- RANK 1 Library fine payment process
- RANK 2 **Payment reconciliation for debited transactions**
- RANK 3 Online examination registration
- RANK 4 Refund policy
- RANK 5 Hostel payment

1. Most relevant? **Rank 2** · Rank 1 ideal? **No**
2. Reranking help? **Yes** · Metadata? `department = Accounts`
3. Give the LLM all 50 chunks? **No**

Same chunks, same model — only **ordering** changed. Today's tools apply at once: **similarity ≠ correctness · reranking · metadata · Top-k discipline.**

CHECK UNDERSTANDING RUN AS TRUE / FALSE — VOTE FIRST, THEN REVEAL

Common Misconceptions — True or False?

1. Highest similarity always means factually correct. **FALSE**

2. Larger chunks are always better. **FALSE**

3. Higher Top-k is always better. **FALSE**

4. Metadata can improve retrieval precision. **TRUE**

5. Reranking occurs before candidate retrieval. **USUALLY FALSE** — in the common two-stage pattern.

6. RAG completely prevents hallucination. **FALSE**

7. Retrieval and generation should be evaluated separately. **TRUE**

8. A correct final answer proves retrieval was good. **NOT NECESSARILY** — the model may have answered from parametric knowledge.

Excellent nuance for students: item 8 — a correct answer can come from the model's memory, not from your retrieval. **Correct output does not certify the pipeline.**

THE BIG PICTURE EVERY STAGE IS A QUALITY GATE

The Complete Retrieval Quality Pipeline



WHERE TODAY'S PARTS LIVE

- PART B — Good chunking — strategy, size, overlap
- PART C — Top-k candidates — the funnel knob
- PART D — Metadata filtering — the relevance + rights signal
- PART E — Reranking — expensive precision, second stage
- PART G — Grounded prompt — instructions + abstention
- PART H — Evaluated answer — retrieval + generation metrics

Read it as a chain of gates: quality is designed at every stage — weak evidence cannot be rescued downstream.

CLOSING FIVE THINGS · ONE FORMULA · NEXT STOP

Five Things to Remember — and What Comes Next

1. **Similarity is not the same as correctness.**
2. **Chunking directly affects retrieval quality.**
3. **More retrieved context is not automatically better.**
4. **Reranking improves candidate ordering** after retrieval.
5. **Evaluate retrieval and generation separately.**

Leave with this sentence: a reliable RAG system does not merely retrieve some text; it retrieves the **right evidence**, ranks it appropriately, supplies **focused context**, and separately evaluates **retrieval and generation**.

LECTURE SUMMARY — RAG QUALITY IS BUILT FROM

Similarity Search · Chunking Strategy · Top-k · Metadata · Reranking · Grounding · Evaluation

RETRIEVAL EVALUATION → "Did we find the right evidence?"

GENERATION EVALUATION → "Did the LLM use that evidence correctly?"

`Retrieve → Filter → Rerank → Ground → Generate → Evaluate`

NEXT · LECTURE 4

Real-World RAG Applications: Healthcare → Finance → Customer Support → Education → Production Design Considerations

"Lecture 3 taught us how to make retrieval trustworthy. Lecture 4 shows what trustworthy retrieval can enable in the real world."